

Hugo Chen

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EXPERIENCE

- Software Engineer Intern** July. 2026 – Sept. 2026
TSMC (Taiwan Semiconductor Manufacturing Company) *Hsinchu, Taiwan*
- Developed an AI assistant capable of analyzing 60,000+ semiconductor machine logs and correlating them with corresponding codebases to identify root causes and accelerate debugging.
 - Collaborated with 9 other interns and managers to engineer an OpenHarness-based multi-agent system with MCP servers, custom tools, and skills.
 - Implemented Azure DevOps CI/CD pipelines to automate deployment of containerized AI services to TSMC's internal testing environment using Docker and Kubernetes.
- Wood Framer/Web Developer** Jun. 2023 – Oct. 2024
Linty Constructions *Vancouver, BC*
- Framed 4 custom residential homes, including an energy-efficient residential project
 - Streamlined coordination by leading English/mandarin communications between clients and contractors.
 - Launched company website which improved online visibility and increased client inquiries
- Bike Mechanic** Jun. 2022 – Jun. 2023
Canadian Tire *Coquitlam, BC*
- Achieved one of the lowest rework and return rate through systematic inspection and tuning
 - Recognized by managers for precision and reliability during peak seasonal demand
 - Advised customers on bike selection, achieving top monthly sales in the store

PROJECTS

- AI Trader** | *Python, PyTorch, Pandas, yFinance* Aug. 2025 – Present
- AI trading system in Python using reinforcement learning (Q-learning & policy gradients) with custom technical indicators (RSI, Stochastic, EMA)
 - Live paper trading using REST API integration and organized using Pandas dataframe.
 - Built comprehensive logging with live JSON profiles and CSV outputs for analysis
- BehindTheETF** | *Next.js, React, TypeScript, PostgreSQL* Jun. 2026 – Present
- Built an ETF analytics platform with recursive fund look-through beyond standard portfolio tools
 - Designed a TypeScript ingestion pipeline for 5 ETF providers and issuer-specific data integrations.
 - Implemented APIs for holdings, sector, country, and overlap exposure across nested ETFs.
- Financial Tracker** | *Java, Swing* Sep. 2025 – Jan 2026
- Created a Java Swing GUI to track income, expenses, and account balances
 - Applied OOD principles to ensure a scalable program architecture
 - Developed JUnit tests to achieve high test coverage and ensure model correctness

EDUCATION

- University of British Columbia** Vancouver, Canada
B.A.Sc. in Electrical Engineering, Minor in Computer Science *2025 – 2029*
- GPA: 89%, Dean's List
- Simon Fraser University** Burnaby, Canada
Concurrent Studies, Non-Degree student while in highschool *2023 – 2025*
- GPA: 3.91/4.33

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Kotlin, JavaScript, HTML/CSS, MATLAB, Assembly, TypeScript, SQL
Others: Microsoft Suite, AutoCAD, Fusion 360, WordPress, Pandas, NumPy, React, Scikit Learn, Docker, Kuberentes, Git
Spoken Languages: Mandarin (Native), Japanese (Advanced), English (Native)